

[Collaborative Docs](#) >

Discussion Agenda 08/10, 08/23, 10/01, 10/29, 11/14, 12/13/2019 (CosmoRun+CGM)

Below is a discussion agenda / live progress report for the Collaboration-wide brainstorming session of the 8th AGORA Workshop @ University of California Santa Cruz (Aug. 9–10, 2019; see the [program](#)).

8th AGORA Workshop participants:

- During this year's workshop, we will have the joint working sessions for Papers "CosmoRun" and "CGM", with code leaders who have carried out the isolated/cosmological runs for the two Papers, and participated in monthly web conferences.

- Everybody should feel free to edit the [meeting notes](#). The content of the meeting note has now been merged into relevant Workspace pages.

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Recent important discussions that require your attention are in red color!

(1st revision: 07/19/2019, 2nd revision 08/10/2019, 3rd revision 08/14/2019, 4th revision 09/25/2019, 5th revision 10/03/2019, **6th revision 12/13/2019**)

[1. Paper Group Progress Report \(as of Aug. 2019\)](#)

[\(1\) Ongoing Projects](#)

	Paper "CosmoRun" + Paper "CGM"	Paper "HighresRun" + Paper "Clumps"																																				
GOALS	• Paper "CosmoRun" : Introduce ~1e12 Ms halo run, emphasis on the calibration process and how it differs from previous attempts like Aquila, discuss basic properties of halos • Paper "CGM" : Study absorption lines, focus on CGM properties that are not resolution–dependent (e.g., high ions)	• Paper "HighresRun" : Run ~1e12 Ms halo at higher resolution (20 pc instead of 80 pc) to study SF clumps at high z, nice resolution study when compared to Paper "CosmoRun" • Paper "Clumps" : Study star–forming clumps, compare with observations																																				
LEADERS	• Cameron Hummels (Caltech) • Santi Roca–Fabrega (UCM)	• Daniel Ceverino (Heidelberg) • Alessandro Lupi (IAP)																																				
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PARTICIPANTS	GEAR	Hausammann, Revaz	RAMSES	Agertz, Roca-Fabrega
	GIZMO	Dong, Hopkins, Lupi, Hummels		(Code leaders as of Nov. 2019)
	RAMSES	Agertz, Roca-Fabrega, Teyssier		– ICs help: Buehlmann, Hahn – Analysis help: Chakrabarti, Kim, Turk, Roca-Fàbrega, Strawn
	(Code leaders as of Nov. 2019) – ICs help: Buehlmann, Hahn, Simpson – Analysis help: Chakrabarti, Fumagalli, Hummels, Kim, Madau, Nagai, Roca-Fabrega, Smith, Turk, Strawn • Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – V (subgrid physics) – IX (characteristics of cosmo. galaxies) – X (outflows)		• Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – XI (high-z galaxies)	
PROGRESS MADE OR BEING MADE	• Visit Paper "CosmoRun"+"CGM" Workspace for the latest updates including issues being faced by the group and the individual code groups, and the Overleaf draft being written. • Code leaders have submitted preliminary datasets for the proposed cosmological run aimed at calibrating the feedback at $z=4$ first. Check out the revised target stellar mass at $z=4$ in the Paper "CosmoRun"+"CGM" Workspace . • Strawn has provided an example script using yt+Trident to produce the absorption lines, HI covering fraction, heat maps on radius – column density plane, etc.		• Visit Paper "HighresRun"+"Clumps" Workspace for the latest updates.	
OTHER LINKS	• Paper Workspace • Trident download and documentation • Trident method paper (Hummels et al. 2016)		• Paper Workspace	

2. Future Science-oriented Paper Ideas

(1) Offshoot Paper Ideas Out of Paper "CosmoRun" (80 pc resolution):

- Morphology: comparison with observation (using e.g., Huertas-Company's tool)
- Emission lines: focus on a few lines that observers will be interested in (using e.g., Barrow's tool which he is willing to share and possibly merge into yt)

(2) Offshoot Paper Ideas Out of Paper "HighresRun" (20 pc resolution):

- Dark matter: previously Science-oriented Working Group VII (pitched in this workshop by Jiang/Freundlich; a DM-only run needed along with the 20 pc resolution "HighresRun"; see also "Core/Cusp" in [this link](#))
- Satellite galaxies: previously Science-oriented Working Group VIII

- Dwarf galaxies: previously Science-oriented Working Group VI (pitched in this workshop by Hausammann; a new run needed with a 10^{10} Ms IC)

(3) Offshoot Paper Ideas Out of Our Paper "Disk" (Isolated Disk Run Paper Previously Published):

- See Paper "Disk-II" in the table down below (Section 4--(8)).
- See also "Secular Evolution" in [this link](#).

(4) How to Relaunch/Reinitiate Science-oriented Working Groups

- Please see Section 3--(3) below to learn more about how we plan to relaunch/reinitiate the science-oriented Working Groups in the 2020 Workshop.

3. More Items to Discuss

(1) AGORA Resources on NERSC:

- We thank the annual effort by Primack and Abel to secure our continuous access to NERSC computing resources. Currently our allocation for the year is 200 TB storage and half a million cpu hours (for analysis job only!). We also concluded that applying for simulation times at NERSC is probably unnecessary.

(2) Planned Public Release of the AGORA Isolated Disk Simulation Data

- We plan to publicly release the simulation data from our published, isolated disk comparison paper (Paper II, or Paper "Disk", or formerly [Paper "4"](#)). By making the data public, the goal is to (1) allow the community to initiate their own new analyses other than the ones presented in our Paper, and (2) give a tool to young students to get involved/interested in AGORA or in galaxy formation simulations in general.

• Therefore, we ask for the permission from the code leaders of Paper II (see the blue-colored names in Paper "Disk-II" in the table down below; Section 4--(8)). If you have any objection, please inform the [Project Coordinator](#) no later than 08/31/2019 (everybody has agreed since!; update on 09/25/2019). Please also share with us any idea on how to store the data so it is most easily accessible.

• Primack is in touch with NERSC to figure out the options they can offer us. We also agree to write a short note with a description of the released data and of the AGORA project to be uploaded to the arXiv. Roca-Fabrega and Kim will write the first draft and will circulate it with all code leaders (update on 10/03/2019).

• Roca-Fabrega has produced the draft which will be circulated to a wider audience soon. Kim is working to post the data on a public space such as Google Drive (link [here](#)). We also have compiled all the links useful for both outsiders ("Quick Links" [here](#)) and for Collaboration members ("Extended Quick Links" [here](#)). We expect to post the note on the arXiv in early 2020 (update on 12/13/2019).

(3) Goal of Next Year's Workshop (2020)

• In the 2020 Workshop, with the completion of Paper "CosmoRun", we aim to relaunch/reinitiate the 9 Science-oriented Working Groups (science topics first identified in 2012–2013; more information in [this link](#)). Due to various changes in the past years, it is expected that the Science-oriented Working Group leadership positions will be reopened for younger postdocs and students to grab in the next year's Workshop.

• We also expect to have a much larger workshop with wider science topics (i.e., not like the technically very focused ones in the past two years). So please mark your calendar already and stay tuned. And if possible, start contributing your time and work to Paper "CosmoRun" and Paper "CGM".

4. Recaps

(1) How to Request an Account on NERSC:

- If you represent a code group but don't have a NERSC user account, please visit [this page](#). Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.

(2) Strategy to Store Simulation Outputs:

- Please visit [this link](#).

(3) AGORA SMBH Prescription for Accretion and Feedback:

- Please visit [this link](#).

(4) Your Own yt Installation on NERSC:

- Please visit [this link](#).

(5) yt-Sunrise Pipeline for AGORA on NERSC / Working Installation of Sunrise as Docker Image:

- Please visit [this link](#).

(6) Mini-workshop Opportunities:

- Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the [yt-AGORA mini-workshop](#) at UCSC in Mar. 2014.

(7) Other Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)
- Collection of softwares relevant to AGORA effort: [here](#)

(8) Previously Launched Projects:

	Paper "Disk-II"	Paper "Metallicity"																				
GOALS	• Compare simulation codes using common disk ICs	• Run ~1e12 Ms halo to study metal distribution																				
LEADERS	• Santi Roca-Fabrega (UCM)																					
PARTICIPANTS	• Code representatives:	• Code representatives:																				
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PROGRESS MADE OR BEING MADE	- Visit Paper's Workspace where the results for disk comparison are compiled (including those presented in our first disk comparison paper – previously known as Paper "4"). The agora-analysis-script repository has been accordingly updated. - Analysis items that were not covered in our first disk comparison paper (see this link) will be considered in the next paper, Paper "Disk-II". - A subset of topics could be discussed in the "Metal Diffusion" or "Secular evolution" paper (see this link).	- Visit Paper's Workspace for the latest updates. - Will piggyback on the planned Paper "CGM" runs and evolve them further down to z=0.										
OTHER LINKS	- Progress report in Paper's Workspace and/or in WG V's Workspace. - yt-AGORA Google group	Paper Workspace										

	Paper "DM-Highres"	Paper "NoSubgrid"																																														
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo) at higher res	• Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)																																														
LEADERS	• Oliver Hahn (OCA, Nice) • Christine Simpson (HITS)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (SDSC)																																														
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PROGRESS MADE OR BEING MADE	• Visit Paper's Workspace where Simpson, Hahn and others have been updating the progress. • We will plan to have 5-code comparison. As of Aug. 2017, 4+ participating groups have completed and submitted the production-level results (~100 pc resolution) to NERSC. • We may focus on the (survival of) substructures. This subset of topics could potentially to be led by Jiang as a smaller paper. • Consistent tree – python interface has been built by Britton Smith (ytree).	• Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.
OTHER LINKS	• Progress report in Paper's Workspace and/or in WG III's Workspace. • MUSIC Google group • ytree documentation	• GRACKLE Google group • GRACKLE download & documentation • GRACKLE method paper (Smith et al. 2017) • AGORA API for GRACKLE by Nick Gnedin

[5. Your Feedback](#)

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!